Constructed from lightweight materials such as wood, canvas, and early metal alloys, first-generation drones had very limited payload-carrying capability. This restricted their use to basic tasks like target practice, as they could not accommodate cameras, sensors, or heavy explosive payloads effectively. As a result, their functionality remained confined to experimental and training purposes.

- *No autonomy or navigation systems*
  First-generation UAVs operated solely under manual control, with no onboard systems for autonomous flight, navigation, or decision-making. They lacked essential instruments such as altimeters, gyroscopes, and compasses, making them heavily dependent on the operator's visual tracking. This absence of automation limited their operational reliability and mission complexity.
- *High crash and failure rates*
  Due to their mechanical simplicity and technological limitations, early drones had high rates of failure and crashes. Many prototypes were lost during testing because of poor stability, signal loss, or engine malfunction. Each crash represented a significant financial and technical setback, as parts had to be manually rebuilt and recalibrated.
- *Environmental and weather sensitivity*
  First-generation drones were extremely vulnerable to weather conditions, including strong winds, humidity, and temperature fluctuations. Their light structures and low engine power made them unsuitable for operation under adverse environmental conditions, further limiting their practical usability.
- *Lack of real-time feedback and data transmission*
  These early drones had no telemetry or real-time feedback systems, preventing operators from monitoring performance parameters like